SC4001 – Part B: Chest X-ray Classification

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Generative AI Declaration:

I used **Codex** to assist with the generation of selected code segments during the development of this assignment. The tool was used primarily to generate initial code templates and to obtain suggestions for implementing specific programming tasks. All AI-generated code was carefully reviewed, modified where necessary, integrated into the overall solution by me, and tested to ensure it met the assignment requirements. I verified the correctness of the generated code through manual inspection, compilation/execution testing, debugging, and comparison with the assignment specifications and relevant course materials. The final submitted work represents my own understanding and effort. I take full responsibility for the accuracy, functionality, and academic integrity of the submitted solution.

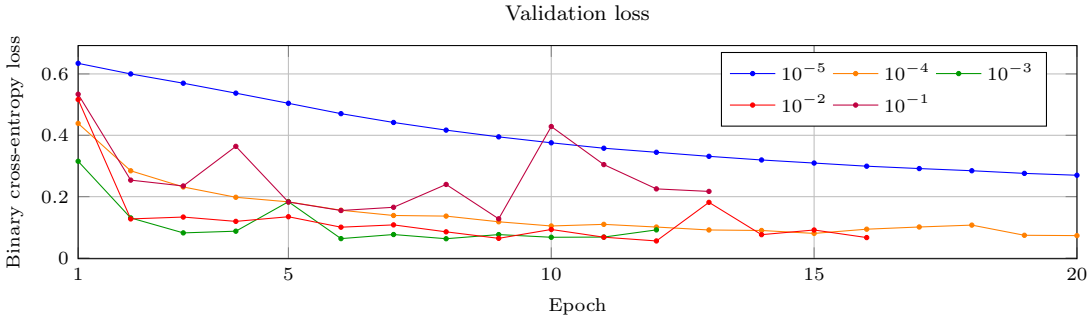
Generative AI Tool Used: Codex

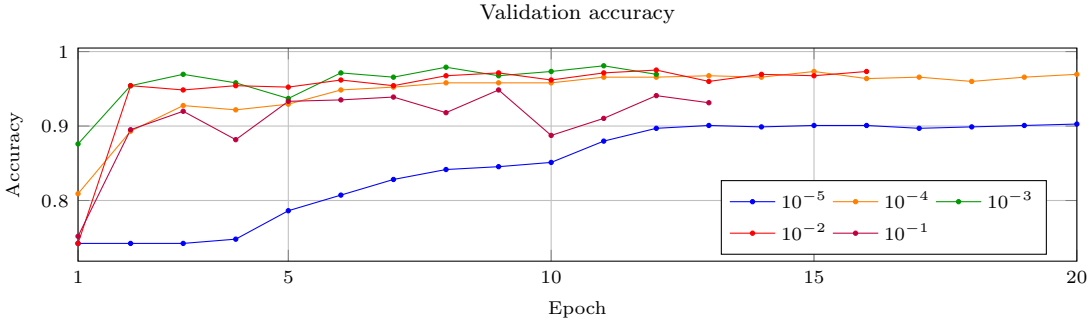
Purpose of Use: Code generation and programming assistance

Verification Method: Manual code review, testing, debugging, and validation against assignment requirements

B1

(a) Validation Loss and Accuracy Curves for Different Learning Rates





(b) Learning Rate Comparison Table

Pneumonia is positive (label 1); the prediction threshold is 0.5. TP, TN, FP, FN denote true positives, true negatives, false positives and false negatives; $N = TP + TN + FP + FN$.

$$\text{Accuracy} = \frac{TP + TN}{N}, \qquad \text{Error} = 1 - \text{Accuracy},$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}.$$

Validation loss is binary cross-entropy, with true label y_i and predicted pneumonia probability p_i:

$$\mathcal{L}_{\text{val}} = -\frac{1}{N} \sum_{i=1}^{N} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

Each row reports the minimum-error checkpoint (ties: lower loss, then earlier epoch). Error, sensitivity and specificity are percentages.

Learning rate	Best epoch	Epochs run	Val. loss	Error (%)	Sens. (%)	Spec. (%)
10^{-5}	20	20	0.2698	9.73	96.14	73.33
10^{-4}	15	20	0.0803	2.67	97.69	96.30
10^{-3}	11	12	0.0683	1.91	97.94	98.52
10^{-2}	12	16	0.0557	2.48	98.20	95.56
10^{-1}	9	13	0.1278	5.15	96.92	88.89

(c) Optimal Learning Rate and Discussion

Optimal learning rate is 10^{-3} with the lowest validation error of 1.91%. The learning curves show that the model with learning rate 10^{-3} converges faster than other learning rates, with the best validation loss and accuracy in most of the epochs. Sensitivity and specificity are also needed to interpret model performance because they reflect the models' ability to detect pneumonia cases and correctly identify healthy subjects, respectively. And the learning rate 10^{-3} has high sensitivity and specificity, proving its effectiveness in accurate detection of both categories.

(d) Test Performance

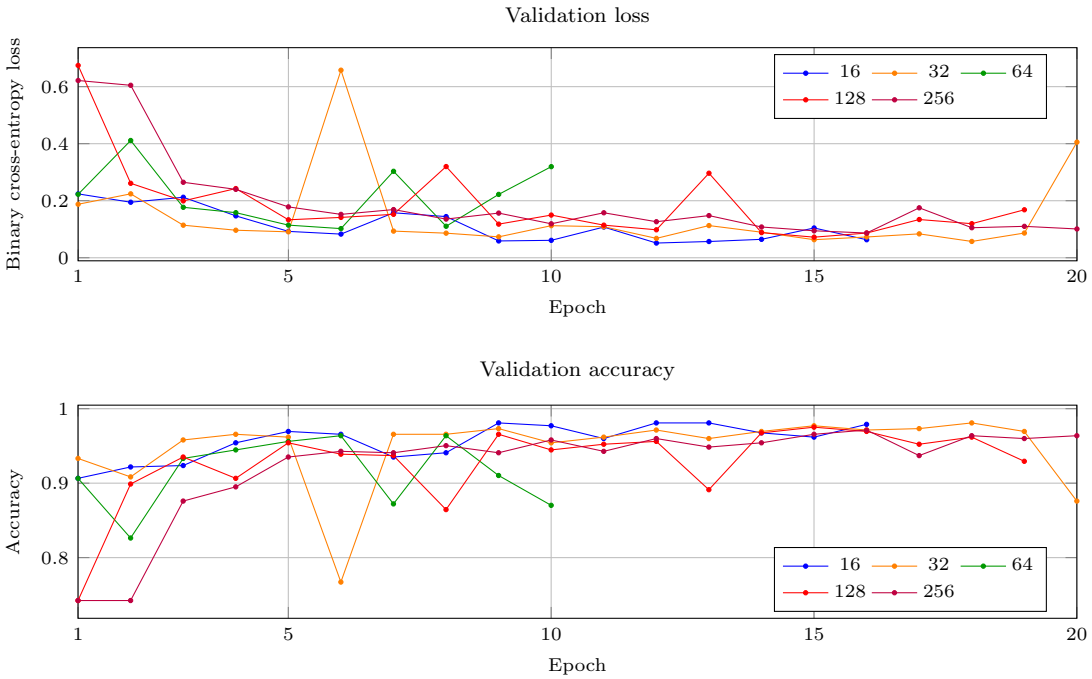
The validation-selected checkpoint (10^{-3}, epoch 11) was evaluated once on 624 test images using training-set normalisation and a threshold of 0.5.

Accuracy (%)	Sensitivity (%)	Specificity (%)
87.02	99.49	66.24

The confusion counts were $TP = 388$, $TN = 155$, $FP = 79$ and $FN = 2$.

B2

(a) Validation Loss and Accuracy Curves for Different Batch Sizes



(b) Batch Size Comparison Table

Metrics follow B1(b). Each row uses the lowest-error checkpoint, with ties resolved by lower validation loss and then earlier epoch.

Batch size	Best epoch	Epochs run	Val. loss	Error (%)	Sens. (%)	Spec. (%)
16	12	16	0.0516	1.91	98.20	97.78
32	18	20	0.0572	1.91	98.97	95.56
64	6	10	0.1022	3.63	97.43	93.33
128	15	19	0.0722	2.48	98.97	93.33
256	16	20	0.0868	2.86	98.20	94.07

(c) Optimal Batch Size and Discussion

Batch sizes 16 and 32 share the lowest validation error of 1.91%. We choose batch size 16 as the optimal one because its validation loss is lower (0.0516 versus 0.0572), and it has a more balanced sensitivity and specificity.

(d) Test Performance

The selected batch-size-16 checkpoint (epoch 12) was evaluated once on 624 test images using training-set normalisation and a threshold of 0.5.

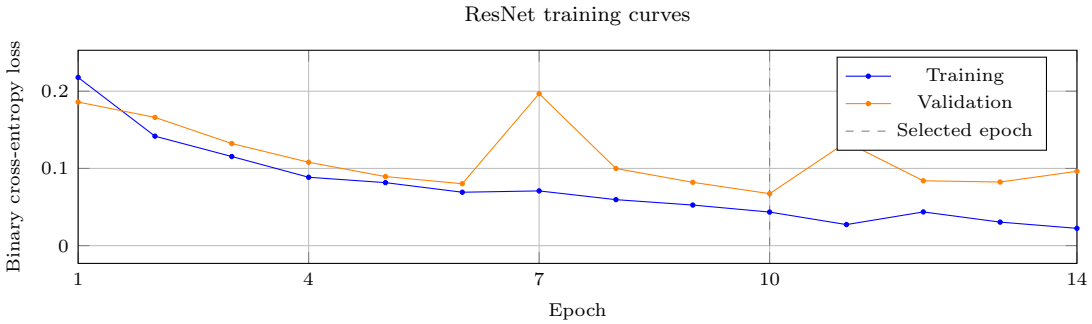
Accuracy (%)	Sensitivity (%)	Specificity (%)
88.62	99.49	70.51

The confusion counts were $TP = 388$, $TN = 165$, $FP = 69$ and $FN = 2$.

B3

(a) Model Performance

The ResNet-like model used learning rate 10^{-3}, batch size 16 and seed 42, with the same preprocessing and early-stopping rules as B1–B2. Training stopped at epoch 14.



The lowest-validation-error checkpoint (epoch 10) was evaluated once on 624 test images with a threshold of 0.5.

Accuracy (%)	Sensitivity (%)	Specificity (%)
87.82	100.00	67.52

The confusion counts were $TP = 390$, $TN = 158$, $FP = 76$ and $FN = 0$.

(b) Comparison and Discussion

Model	Acc. (%)	Sens. (%)	Spec. (%)	Error (%)	Parameters	MFLOPs
AlexNet-like	87.02	99.49	66.24	12.98	99,569	39.06
VGG-like	88.62	99.49	70.51	11.38	18,185	19.46
ResNet-like	87.82	100.00	67.52	12.18	77,169	25.30

Parameters include all trainable weights and biases. FLOPs measure convolutional and fully connected computation per $1 \times 64 \times 64$ image, counting each multiply–accumulate as two operations.

AlexNet-like: Its 5×5 kernels provide larger local receptive fields. The fully connected head contributes to the highest parameter count, and the model has the highest FLOPs and lowest test accuracy and specificity.

VGG-like: Stacked 3×3 convolutions add nonlinear transformations, and global average pooling keeps the classifier compact. It achieves the highest accuracy and specificity with the fewest parameters and FLOPs. Global pooling trades spatial detail for compactness; the model produces 69 false positives.

ResNet-like: Residual shortcuts provide direct gradient paths that can ease optimisation. The model detects all 390 pneumonia cases and produces 76 false positives. VGG offers higher accuracy and specificity at lower computational cost.

Model	Pros	Cons
AlexNet-like	Larger local receptive fields from 5×5 kernels; high sensitivity (99.49%).	Highest parameter count and FLOPs; lowest specificity (66.24%).
VGG-like	Highest accuracy (88.62%) and specificity (70.51%); lowest parameter count and FLOPs.	Global pooling compresses spatial detail; 69 false positives.
ResNet-like	Residual shortcuts support gradient flow; detects all 390 pneumonia cases.	More computation than VGG; 76 false positives and lower specificity (67.52%).

B4

(a) X-ray Selection and Grad-CAM Implementation

(b) Grad-CAM Overlays

(c) Comparison and Interpretation